Project Approval Form

Course: B. Tech (CSE) **Group No.:189** **Session: 2026-27**

Project Title: **CodeONE —Consolidate Your Competitive Programming Journey**

Project Group Details							
#	University Roll No.	Sec	Class Roll No.	Student's Name	CPI (Upto VI Sem))	Mob. No.	Sign
1.	2315000557	4F	17	Ayush Raj	7.7	8709863406	
2.	2315000506	4E	20	Astha Goswami	7.85	7668781756	
3.	2315000790	4G	21	Divyanshi Kulshrestha	7.6	8439495593	
4.	2315000401	4G	7	Anvesha Singh	7.66	9634074545	
5.	2315000994	4G	27	Ishendra Rai	7.6	9305859379	

Supervisor's Details		
Name	**Designation**	**Contact No.**

ABSTRACT

CodeONE is a full-stack competitive programming analytics platform designed to provide a unified performance intelligence layer for developers, students, and interview aspirants across multiple coding platforms. Built on an API-first modular monolith architecture utilizing React (Vite), Node.js, Express, PostgreSQL, Prisma ORM, and optional Redis caching, it consolidates fragmented user metrics into a single analytics workspace. Through an isolated Platform Adapter Layer, CodeONE prioritizes official and permitted data sources—starting with Codeforces, alongside AtCoder (via AtCoder Problems), Codewars, CLIST.by contest ingestion, and a manual data import fallback for scraping-restricted platforms such as LeetCode and CodeChef—without ever collecting external platform passwords or session cookies. Key capabilities include unified global streak tracking and 365-day activity heatmaps derived from timestamped submissions, per-platform rating trajectories, problem state management, and rule-based topic weakness detection. To protect user telemetry and prevent API abuse, CodeONE enforces a defense-in-depth security architecture featuring Argon2id/bcrypt password hashing, verified Google/GitHub OAuth 2.0, short-lived JWT access tokens (15-minute expiry) paired with server-tracked rotating refresh tokens and token-family reuse detection, default-deny four-role RBAC (User, Moderator, Admin, Auditor) with strict IDOR protection, public-handle ownership verification, Joi schema validation, and append-only audit logging. Ultimately, CodeONE functions as both a consistency engine for competitive programmers and a verifiable model for secure, policy-compliant web architecture.

Supervisor's Sign: _____ Date: _____

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Panel's Suggestions/Remarks:_____

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Details of Project Approval Panel			
#	Panel Member's Name	Approved (Yes/No)	Sign
1.			
2.			
3.			

Date: (**Signature of Project In-charge**)